MD. ATAUR RAHMAN SIKDAR

Email: arsikdarbtge@std.iu.ac.bd | **Phone:** +880 1309 294 988

LinkedIn | Google Scholar | ResearchGate | GitHub | ORCID

EDUCATION

Islamic University Bangladesh *Kushtia-7003, Bangladesh*

B.Sc. in Biotechnology and Genetic Engineering, Faculty of Biological Sciences *Expected August 2027*

- CGPA: 3.75/4.00 (averaged over 6 completed semesters)
- University Merit Scholarship, awarded for 2 consecutive years *Feb 2024 - Feb 2026*
- Relevant Coursework: Microbiology, Biochemistry, Molecular Biology, Genetics, Immunology, Genetic Engineering and Bio-economics

RESEARCH INTERESTS

- Bacteriophage biology and genomics, phage therapy, phage-antibiotic synergy
- Host-phage interactions, comparative and pan-genomics, and gut microbiota dynamics in metabolic disease.

RESEARCH EXPERIENCE

Bacteriophage Biology and Genomics Lab, Islamic University Bangladesh *Kushtia-7003, Bangladesh*

Research Assistant, supervised by Dr. Sudhangshu Kumar Biswas (Professor) *Jan 2026 - Present*

- Currently investigating bacteriophage-mediated restoration of antibiotic susceptibility in drug-resistant uropathogens, alongside studying phage-resistance islands in *V. cholerae* linked to co-evolution with ICP1 lineages.
- Preparing a manuscript titled "Lineage-Associated His-box Motifs Reveal Receptor-Specific Docking in Bacteriophage Tail Fibers: A Computational Approach."

Centre for Genetic Counselling, Education and Research *Dhanmondi, Dhaka-1205*

Research Intern, supervised by Dr. Mustak Ibn Aiub (Associate Professor) *Mar 2026 - Present*

- Conducting integrative genomic and transcriptomic analysis linking host metabolic stress to butyrate-producer bacterial depletion in Type 2 Diabetes Mellitus.

JOURNAL PUBLICATIONS

- **Sikdar, Md Ataur Rahman**, Md Abidur Rahman, Umma Habiba Hafsa, Hossain Md Faruquee, and Sudhangshu Kumar. "BactoFlow: An Automated and Reproducible Pipeline for High-Resolution Bacterial Genomics." Integrative Bioscience Nexus 1(2) (2026).
- Sheikh, Md Kamrul, Md Abidur Rahman, **Md Ataur Rahman Sikdar**, Md Rasheduzzaman, Sumaya Shahin, Umma Habiba, N. M. Hafsa, Hossain Md Faruquee, and Sudhangshu Kumar Biswas. "Bacteriophage-Mediated Control of Extensively Drug-Resistant (XDR) Uropathogens." Integrative Bioscience Nexus 1(2) (2026).

CONFERENCE PRESENTATIONS

- **Sikdar, Md. Ataur Rahman**, Umma Habiba Hafsa, Md. Abidur Rahman, Md. Mehedi Hasan Sumon, Hossain Md. Faruquee, and Sudhangshu Kumar Biswas. "Phage-Resistance Islands (PLEs) in *Vibrio cholerae*: Conserved Domains Underlying Functional Mimicry and Co-evolution with ICP1 Phage Lineages." International Young Biotechnologists Congress, University of Dhaka (2026).
- Rasheduzzaman, Md**.**, Md. Abidur Rahman, **Md. Ataur Rahman Sikdar**, Md. Mehedi Hasan Sumon, and Sudhangshu Kumar Biswas. "Restoring Antibiotic Susceptibility in Resistant *Escherichia coli* via Bacteriophage Intervention." International Scientific Symposium 2026, Islamic University (2026).
- **Sikdar, Md. Ataur Rahman**, Md. Mehedi Hasan Sumon, Md. Bakibillah, Md. Abidur Rahman, Md. Ibrahim, Sabbir Ahmed, Sezanur Rahman, and Sudhangshu Kumar Biswas. "Uncovering His-Box Motif Dynamics in Bacteriophage Tail Fibers: Insights into Host Specificity and Evolutionary Flexibility via Horizontal Gene Transfer." 1st International Conference on Life Science, Health, and Biotechnology (LifeTech), Jashore University of Science and Technology (2026).

MANUSCRIPTS UNDER REVIEW

- Sultana, Mst. Atia, **Md. Ataur Rahman Sikdar**, Md. Mehedi Hasan Sumon, Joytun Nahar, Md. Johorul Islam, Khandker Md. Faisal Alam, Sezanur Rahman, Mohammad Jubair, Mustafizur Rahman, Sudhangshu Kumar Biswas, and Dipak Kumar Paul. "Revealing *Lysinibacillus sphaericus* BBGL-01, a Human Gut–Derived Isolate, as a Promising Probiotic Candidate." *Journal of Genetic Engineering and Biotechnology* (Elsevier), Ms. No. JGEBIO-D-26-00264, Under Review.

- Khatun, Mst. Roma, Mst. Atia Sultana, Sadia Farzana, **Md. Ataur Rahman Sikdar**, N M Yousuf Ivn Ashraf, Md. Rasheduzzaman, Md. Johorul Islam, Md. Toufiq Elahi, Khandker Md. Faisal Alam, Sudhangshu Kumar Biswas, and Dipak Kumar Paul. "Molecular Identification and Antibiotic Resistance of ICU Pathogens with Rare Clinical Detection of Enterobacter hormaechei in Bangladesh." New Microbes and New Infections (Elsevier), Ms. No. NMNI-D-26-00425, Under Review.

SKILLS

- **Wet-Lab:** Bacteriophage isolation & purification, pH & Thermal Stability Test, One-step Growth Curve Assay, bacteria culture, biochemical assays, antibiotic susceptibility testing, DNA extraction/purification/quantification, PCR, gel electrophoresis, UV spectrophotometry
- **Dry-Lab:** NGS & Sanger Data Analysis; Bacteriophage Genomics (Host Prediction, Prophage Identification, Genome Annotation, Lifestyle/Taxonomy Classification, Comparative Genomics, Phylogenetics); Bacterial Genomics (AMR & Virulence Profiling, MLST, Pangenomics); Differential Gene Expression Analysis; Protein Structure Prediction & Molecular Docking
- **Programming:** Python, R, Bash scripting (beginner level)
- **Tools:** Linux (Ubuntu), RStudio, UGENE, Chromas, PyMOL, EndNote, Zotero, Canva
- **Languages:** Bengali (Native), English (Fluent)

AWARDS AND ACHIEVEMENTS

- **Champion** *May 2026*
  Quiz Competition (Workshop on Metagenomics: Application in Public Health, Environmental and Clinical Microbiology) ASM Student Chapter IU, Islamic University Bangladesh
- **Champion** *Jan 2026*
  World Cancer Day Quiz Competition, Cancer Care and Research Trust Bangladesh
- **Awardee (6th Position)** *Aug 2025*
  Rufaida BioMeds Poster Competition — "Comprehensive Workflow for Bacterial Genome Analysis: From SRA Raw Reads to Reference Mapping and Annotation"
- **People's Choice Award** *Feb 2025*
  Poster Presentation, 2nd ISCB RSG Bangladesh CompBio Symposium — "Exploring His-Box Dynamics in Bacteriophage Tail Fibers: Insights into Host Specificity and Evolutionary Flexibility"

WORKSHOPS AND TRAININGS

- Metagenomics: Application in Public Health, Environmental and Clinical Microbiology (1 day) *May 2026*
  ASM Student Chapter IU, Islamic University Bangladesh
- Cancer Bioinformatics: From Beginning to Publication (15 days) *Mar 2026*
  Centre for Genetic Counselling, Education and Research
- Next-Generation Genomics and Drug Design: Hands-On Training (2 days) *Nov 2024*
  American Society for Microbiology Student Chapter, Islamic University
- Python Programming Boot-Camp: Beginner to Intermediate (1 month) *Nov 2023*
  Islamic University IEEE Student Branch

LEADERSHIP EXPERIENCES

- **Executive Member** *Feb 2024 - Jun 2026*
  Network of Young Biotechnologists of Bangladesh (NYBB)
- **Vice Director of Finance** *May 2024 - Dec 2025*
  International Association of Students in Agricultural and Related Sciences (IAAS) Bangladesh - Islamic University Chapter
- **Campus Coordinator** *May 2024 - May 2025*
  Dawn of Bioinformatics Limited
- **Joint Secretary** *Mar 2024 - Feb 2025*
  American Society for Microbiology Student Chapter of Islamic University (ASMIU)

REFERENCES

- **Dr. Sudhangshu Kumar Biswas**, Professor, Bacteriophage Biology and Genomics Lab, Dept. of Biotechnology and Genetic Engineering, Faculty of Biological Sciences, Islamic University, Kushtia-7003, Bangladesh | Email: skbiswas@btge.iu.ac.bd, shu_genetics@yahoo.com | Phone: +880 1718 626 225